

2022 Ph H2 Q8

(a)

State the name of the fundamental force involved in beta decay

✔ The weak nuclear force.

(b)

Name the type of beta decay shown in this reaction

Given decay:

$${}^0_{-1}e + \bar{\nu}_e$$

✔ This is **beta-minus decay** (β^-) — a **neutron** decays into a **proton**, releasing an **electron** and **antineutrino**.

(c)

Explain why charge is conserved in this reaction

✔ In beta-minus decay:

- A **neutron** (charge 0) changes into a **proton** (charge +1),
- An **electron** (charge −1) is emitted,
- And an **antineutrino** (charge 0) is also emitted.

$$0 = (+1) + (-1) + 0$$

So, **total charge before = total charge after:**

✔ Charge is conserved.

🧠 Revision Tips

- The **weak force** is responsible for **beta decay**, a process that changes quarks and types of particles
- In **beta-minus decay**, a **down quark** becomes an **up quark**: neutron → proton
- Always check conservation laws: charge, energy, lepton number, etc.
- Antineutrinos and neutrinos are neutral, light, and hard to detect but **essential for conserving lepton number**